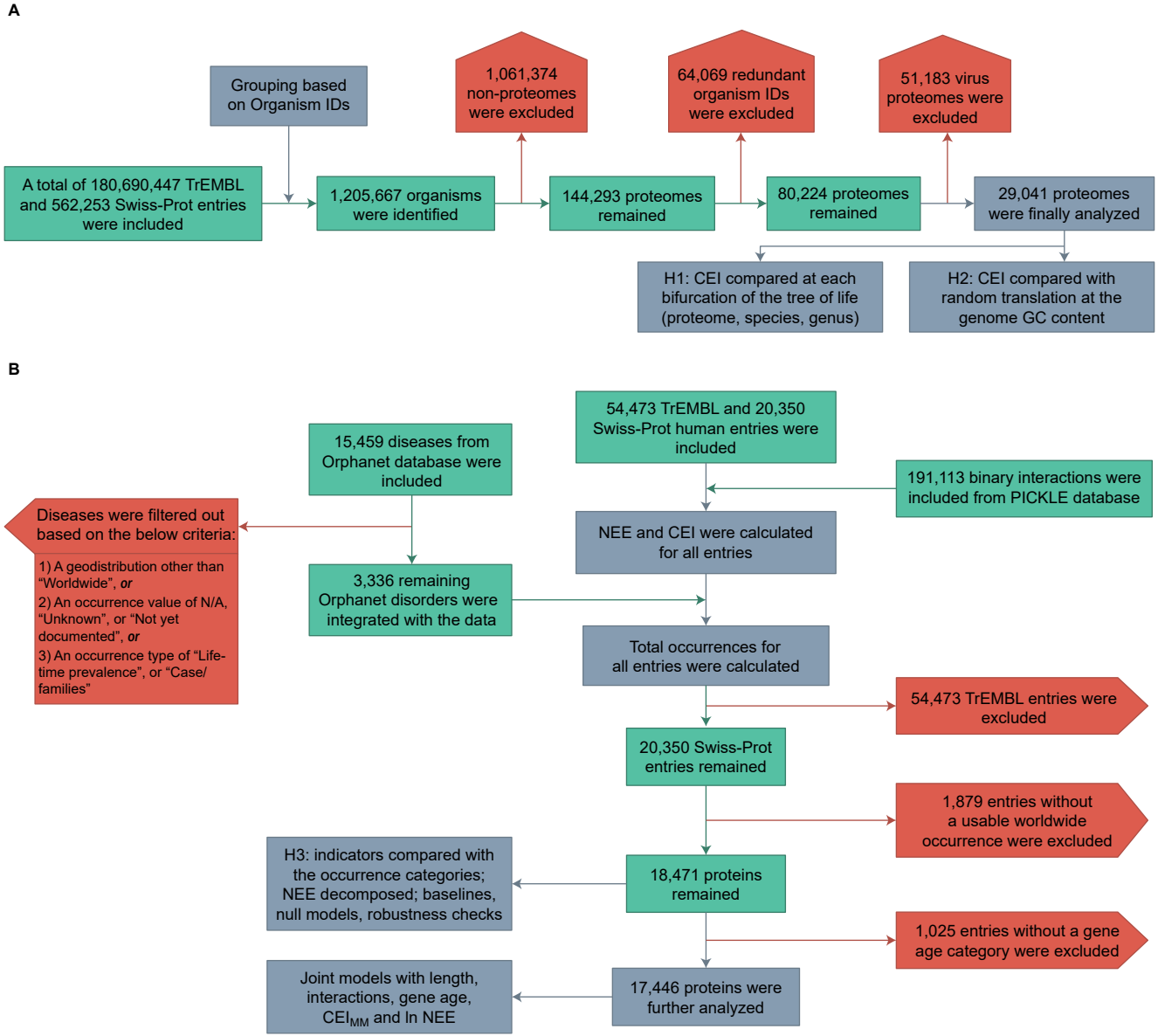
A
Grouping based on Organism IDs
1,061,374 non-proteomes were excluded
64,069 redundant organism IDs were excluded
51,183 virus proteomes were excluded
A total of 180,690,447 TrEMBL and 562,253 Swiss-Prot entries were included
1,205,667 organisms were identified
144,293 proteomes remained
80,224 proteomes remained
29,041 proteomes were finally analyzed
H1: CEI compared at each bifurcation of the tree of life (proteome, species, genus)
H2: CEI compared with random translation at the genome GC content
B
54,473 TrEMBL and 20,350 Swiss-Prot human entries were included
15,459 diseases from Orphanet database were included
191,113 binary interactions were included from PICKLE database
Diseases were filtered out based on the below criteria:
1) A geodistribution other than "Worldwide", or
2) An occurrence value of N/A, "Unknown", or "Not yet documented", or
3) An occurrence type of "Life-time prevalence", or "Case/ families"
NEE and CEI were calculated for all entries
3,336 remaining Orphanet disorders were integrated with the data
Total occurrences for all entries were calculated
54,473 TrEMBL entries were excluded
20,350 Swiss-Prot entries remained
1,879 entries without a usable worldwide occurrence were excluded
H3: indicators compared with the occurrence categories; NEE decomposed; baselines, null models, robustness checks
18,471 proteins remained
1,025 entries without a gene age category were excluded
Joint models with length, interactions, gene age, CEI$_{MM}$ and ln NEE
17,446 proteins were further analyzed